Symmetries in Reversible Programming

Jacek Karwowski

October 25, 2022

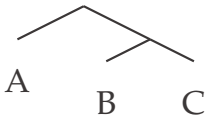
$$A + (B + C) \overset{?}{\longleftrightarrow} C + (B + A)$$

Example - formally

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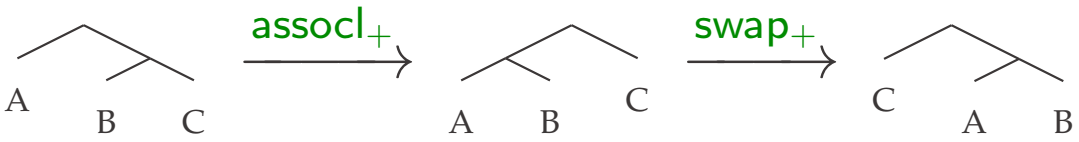
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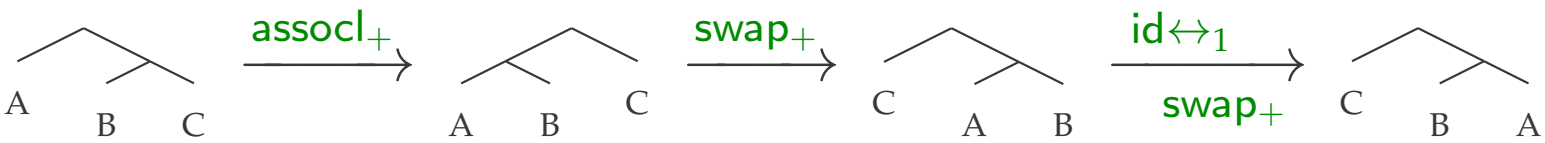
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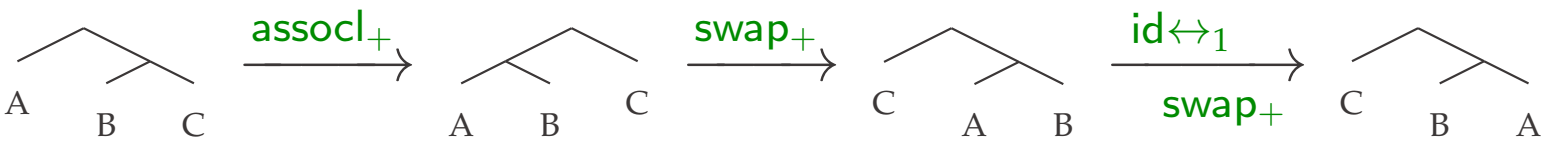
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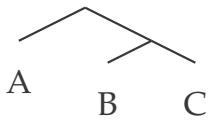
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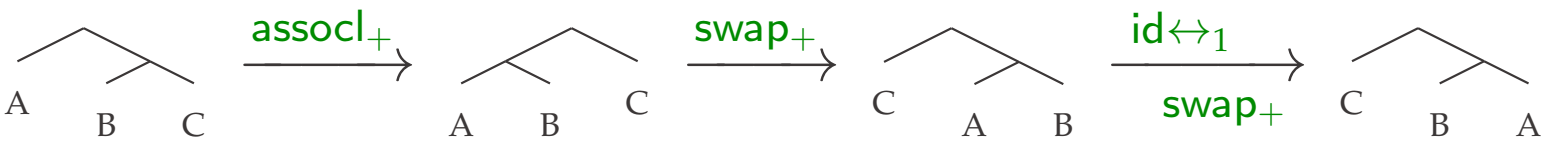
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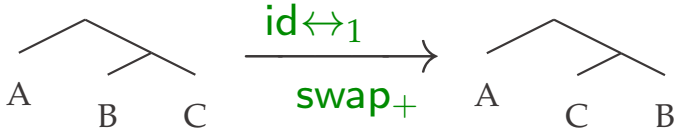




Example - formally

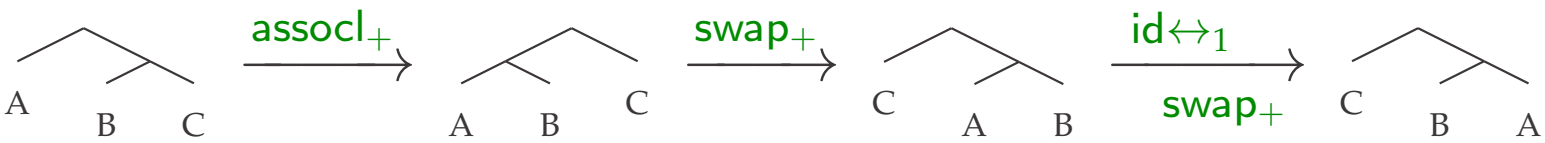
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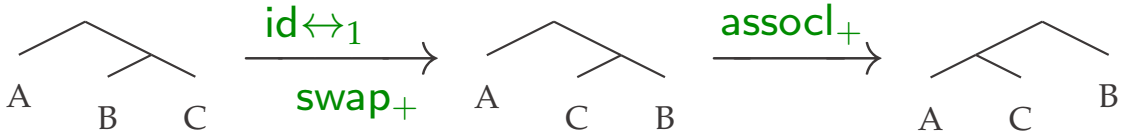




Example - formally

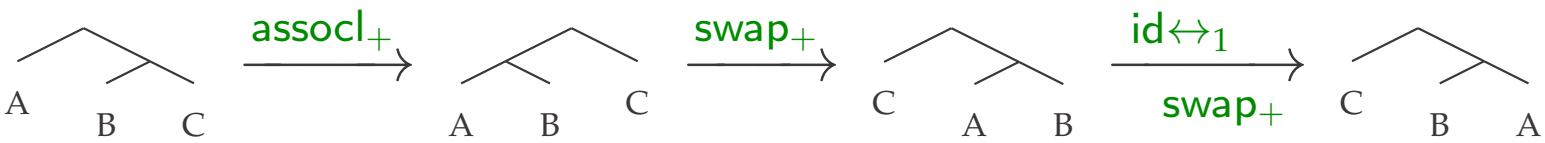
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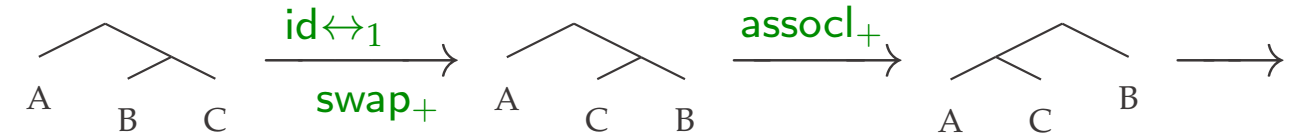


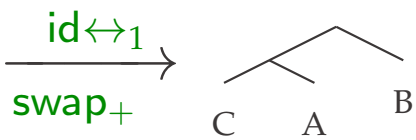


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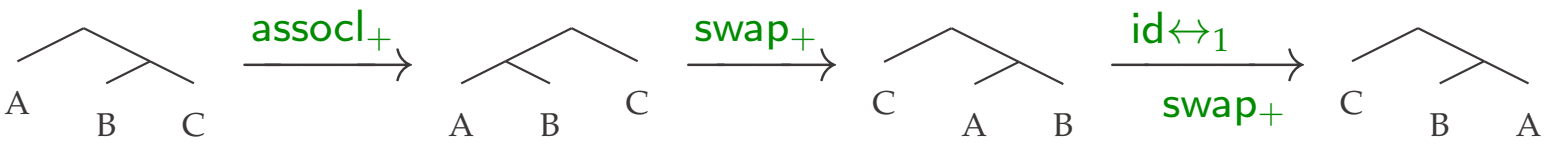


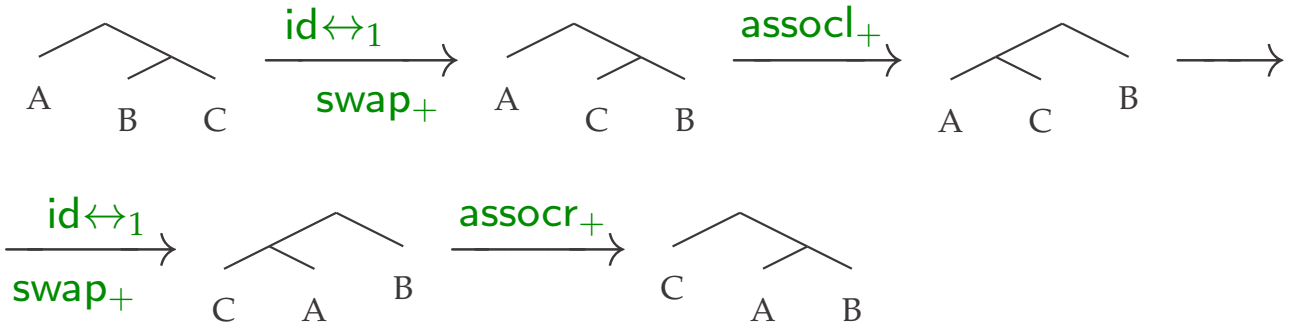




Example - formally

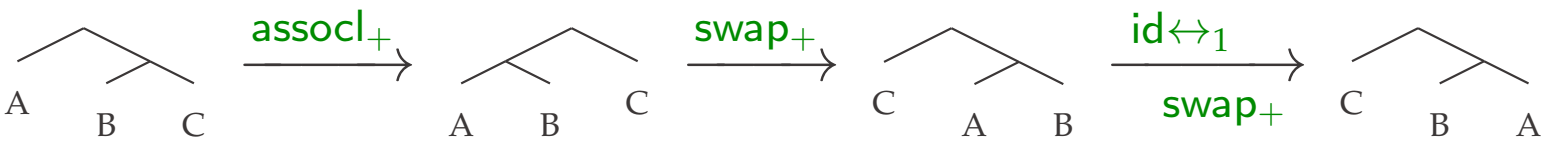
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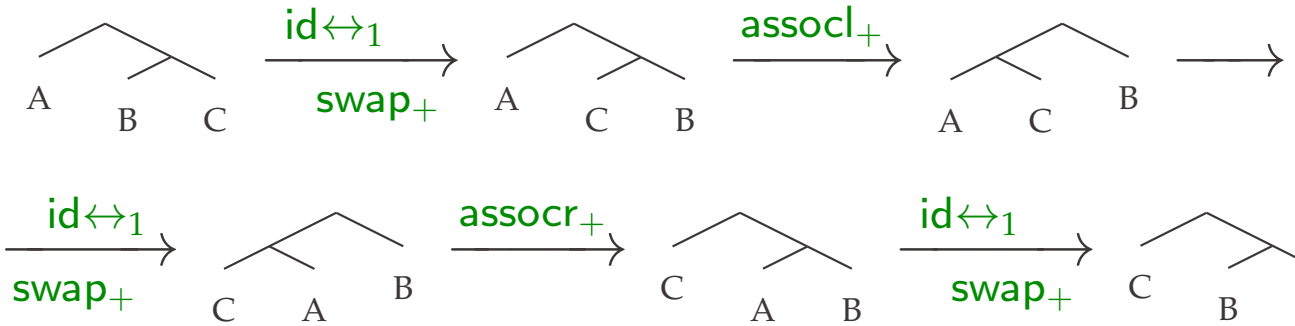




Example - formally

$$A + (B + C) \overset{?}{\leftrightarrow} C + (B + A)$$





Π language - syntax

$$A,\ B,\ C, \ldots := \mathbf{0},\ \mathbf{1},\ A + B,\ A \times B \qquad x,\ y, \ldots := \star,\ \mathbf{inl}\ x,\ \mathbf{inr}\ x,\ (x, y)$$

Types and values

$$\begin{array}{rrcll}
\mathsf{id}{\leftrightarrow_1}: & A & \leftrightarrow_1 & A & : \mathsf{id}{\leftrightarrow_1} \\
\\
\mathsf{unite}_+\mathsf{l}: & \mathbf{0} + A & \leftrightarrow_1 & A & : \mathsf{uniti}_+\mathsf{l} \\
\mathsf{swap}_+: & A + B & \leftrightarrow_1 & B + A & : \mathsf{swap}_+ \\
\mathsf{assocl}_+: & A + (B + C) & \leftrightarrow_1 & (A + B) + C & : \mathsf{assocr}_+ \\
\\
\mathsf{unite}\star\mathsf{l}: & \mathbf{1} \times A & \leftrightarrow_1 & A & : \mathsf{uniti}\star\mathsf{l} \\
\mathsf{swap}\star: & A \times B & \leftrightarrow_1 & B \times A & : \mathsf{swap}\star \\
\mathsf{assocl}\star: & A \times (B \times C) & \leftrightarrow_1 & (A \times B) \times C & : \mathsf{assocr}\star \\
\\
\mathsf{absorbr}: & \mathbf{0} \times A & \leftrightarrow_1 & \mathbf{0} & : \mathsf{factorzl} \\
\mathsf{dist}: & (A + B) \times C & \leftrightarrow_1 & (A \times C) + (B \times C) & : \mathsf{factor}
\end{array}$$

$$\frac{\vdash c_1 : A \leftrightarrow_1 B \quad \vdash c_2 : B \leftrightarrow_1 C}{\vdash c_1 \fatsemi c_2 : A \leftrightarrow_1 C} \qquad \frac{\vdash c_1 : A \leftrightarrow_1 B \quad \vdash c_2 : C \leftrightarrow_1 D}{\vdash c_1 \oplus c_2 : A + C \leftrightarrow_1 B + D} \qquad \frac{\vdash c_1 : A \leftrightarrow_1 B \quad \vdash c_2 : C \leftrightarrow_1 D}{\vdash c_1 \oplus c_2 : A + C \leftrightarrow_1 B + D}$$

Combinators

Example - formally

Two (equivallent) transformations implemented in Π:

$$p_1, p_2 : \mathbf{1}_A + (\mathbf{1}_B + \mathbf{1}_C) \leftrightarrow \mathbf{1}_C + (\mathbf{1}_B + \mathbf{1}_A)$$
$$p_1 = \mathsf{assocl}_+ \circledcirc (\mathsf{swap}_+ \circledcirc (\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+))$$
$$p_2 = (\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+) \circledcirc \mathsf{assocl}_+ \circledcirc (\mathsf{swap}_+ \oplus \mathsf{id}\!\leftrightarrow_1) \circledcirc \mathsf{assocr}_+ \circledcirc (\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+)$$

Example - formally

Two (equivallent) transformations implemented in Π:

$$p_1, p_2 : \mathbf{1}_A + (\mathbf{1}_B + \mathbf{1}_C) \leftrightarrow \mathbf{1}_C + (\mathbf{1}_B + \mathbf{1}_A)$$
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$$q \quad : \mathbf{1}_A + (\mathbf{1}_B + \mathbf{1}_C) \leftrightarrow \mathbf{1}_A + (\mathbf{1}_B + \mathbf{1}_C)$$
$$q = \mathsf{id}\!\leftrightarrow_1$$

Example - formally

Two (equivallent) transformations implemented in Π:

$$p_1, p_2 : \mathbf{1} + (\mathbf{1} + \mathbf{1}) \leftrightarrow \mathbf{1} + (\mathbf{1} + \mathbf{1})$$
$$p_1 = \mathsf{assocl}_+ \circledcirc (\mathsf{swap}_+ \circledcirc (\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+))$$
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Example - formally

Two (equivallent) transformations implemented in Π:

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$$q : \mathbf{1} + (\mathbf{1} + \mathbf{1}) \leftrightarrow \mathbf{1} + (\mathbf{1} + \mathbf{1})$$
$$q = \mathsf{id}\!\leftrightarrow_1$$

$$p_1(\mathbf{inl}(\star)) = \mathbf{inr}(\mathbf{inr}(\star)) = p_2(\mathbf{inl}(\star))$$
$$q\ (\mathbf{inl}(\star)) = \mathbf{inl}(\star)$$

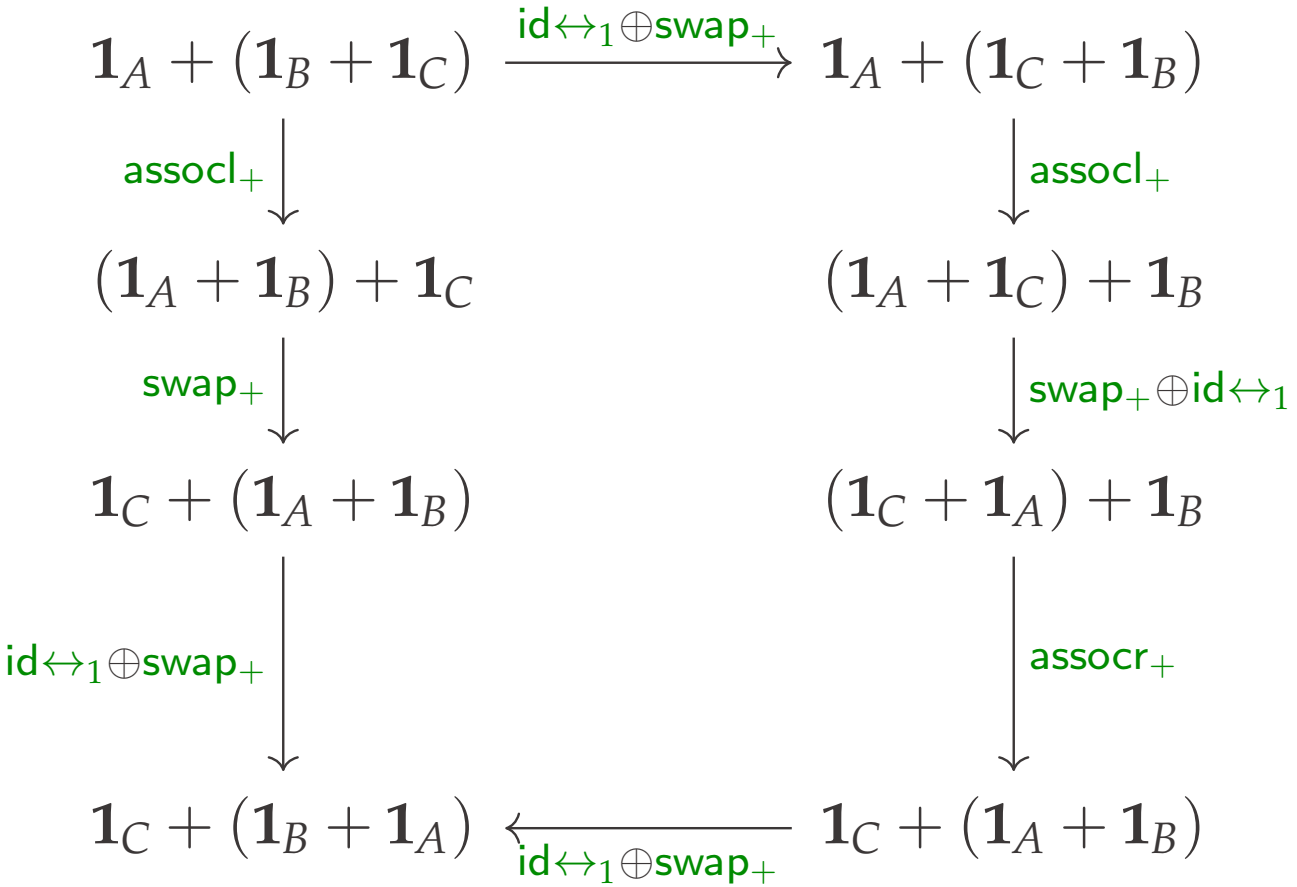
Π - level 2

$$
\begin{array}{rrcll}
\mathsf{id}{\leftrightarrow}_2 : & c & \leftrightarrow_2 & c & : \mathsf{id}{\leftrightarrow}_2 \\
\mathsf{id}{\leftrightarrow}{\oplus}\mathsf{id}{\leftrightarrow} : & \mathsf{id}{\leftrightarrow}_1 \oplus \mathsf{id}{\leftrightarrow}_1 & \leftrightarrow_2 & \mathsf{id}{\leftrightarrow}_1 & : \mathsf{id}{\leftrightarrow}{\oplus}\mathsf{id}{\leftrightarrow}^{-} \\
\mathsf{idl}{\circledcirc}\mathsf{l} : & \mathsf{id}{\leftrightarrow}_1 \circledcirc c & \leftrightarrow_2 & c & : \mathsf{idl}{\circledcirc}\mathsf{l} \\
\mathsf{swap}_+{}^2 : & \mathsf{swap}_+ \circledcirc \mathsf{swap}_+ & \leftrightarrow_2 & \mathsf{id}{\leftrightarrow}_1 & : \mathsf{swap}_+{}^{-2} \\
\mathsf{cassocl}{\circledcirc}\mathsf{r}_+ : & \mathsf{assocl}_+ \circledcirc \mathsf{assocr}_+ & \leftrightarrow_2 & \mathsf{id}{\leftrightarrow}_1 & : \mathsf{eassocl}{\circledcirc}\mathsf{r}_+ \\
\mathsf{assoc}{\circledcirc}\mathsf{l} : & (c_1 \circledcirc c_2) \circledcirc c_3 & \leftrightarrow_2 & c_1 \circledcirc (c_2 \circledcirc c_3) & : \mathsf{assoc}{\circledcirc}\mathsf{l} \\
\mathsf{hom}{\oplus}{\circledcirc} : & (c_1 \oplus c_2) \circledcirc (c_1 \oplus c_2) & \leftrightarrow_2 & (c_1 \circledcirc c_2) \oplus (c_1 \circledcirc c_2) & : \mathsf{hom}{\circledcirc}{\oplus}
\end{array}
$$

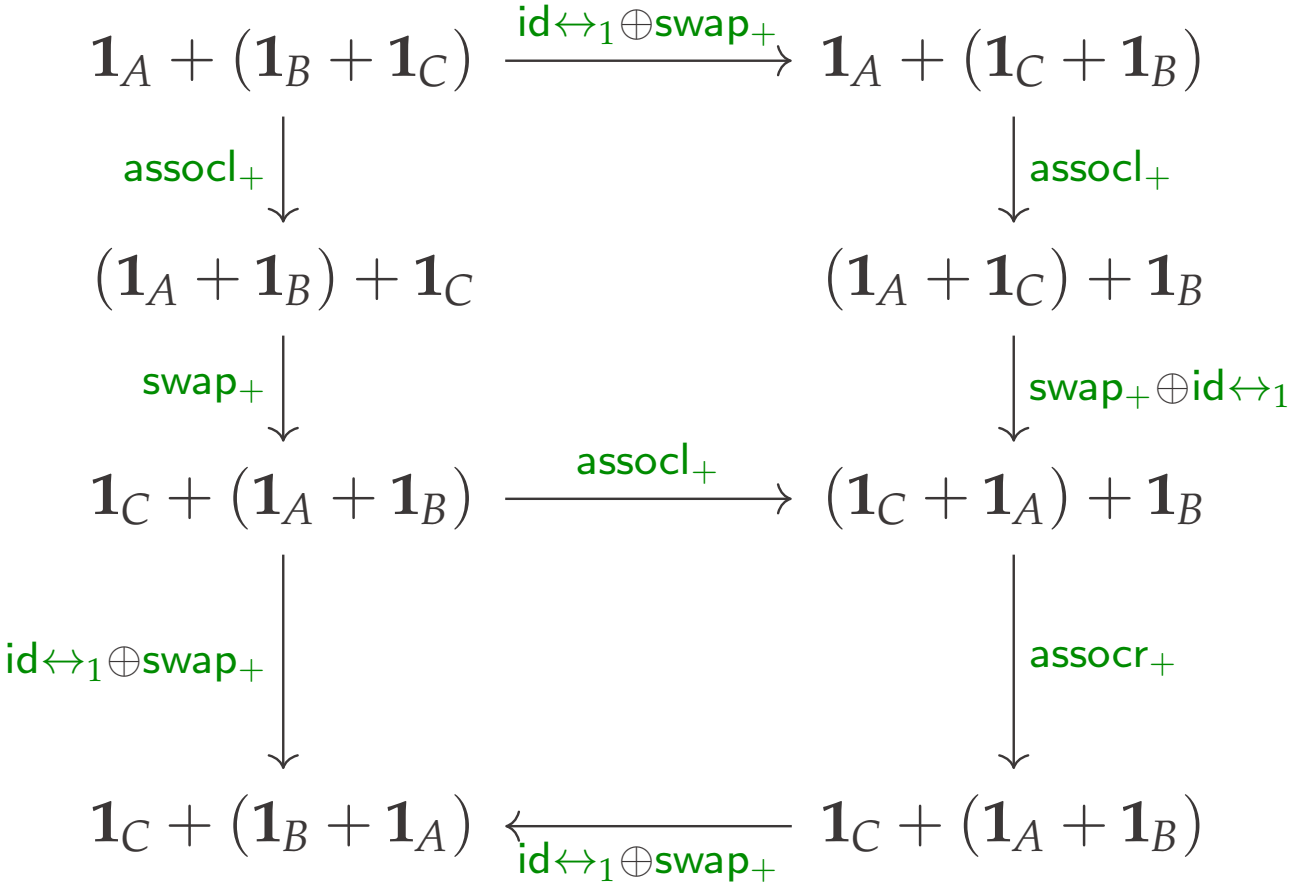
$$\frac{\vdash \alpha : c_1 \leftrightarrow_2 c_2 \quad \vdash \beta : c_2 \leftrightarrow_2 c_3}{\vdash \alpha \blacksquare \beta : c_1 \leftrightarrow_2 c_3} \qquad \frac{\vdash \alpha : c_1 \leftrightarrow_2 c_2 \quad \vdash \beta : c_3 \leftrightarrow_2 c_4}{\vdash \alpha \boxdot \beta : c_1 \oplus c_2 \leftrightarrow_2 c_3 \oplus c_4}$$

Examples of 2-combinators

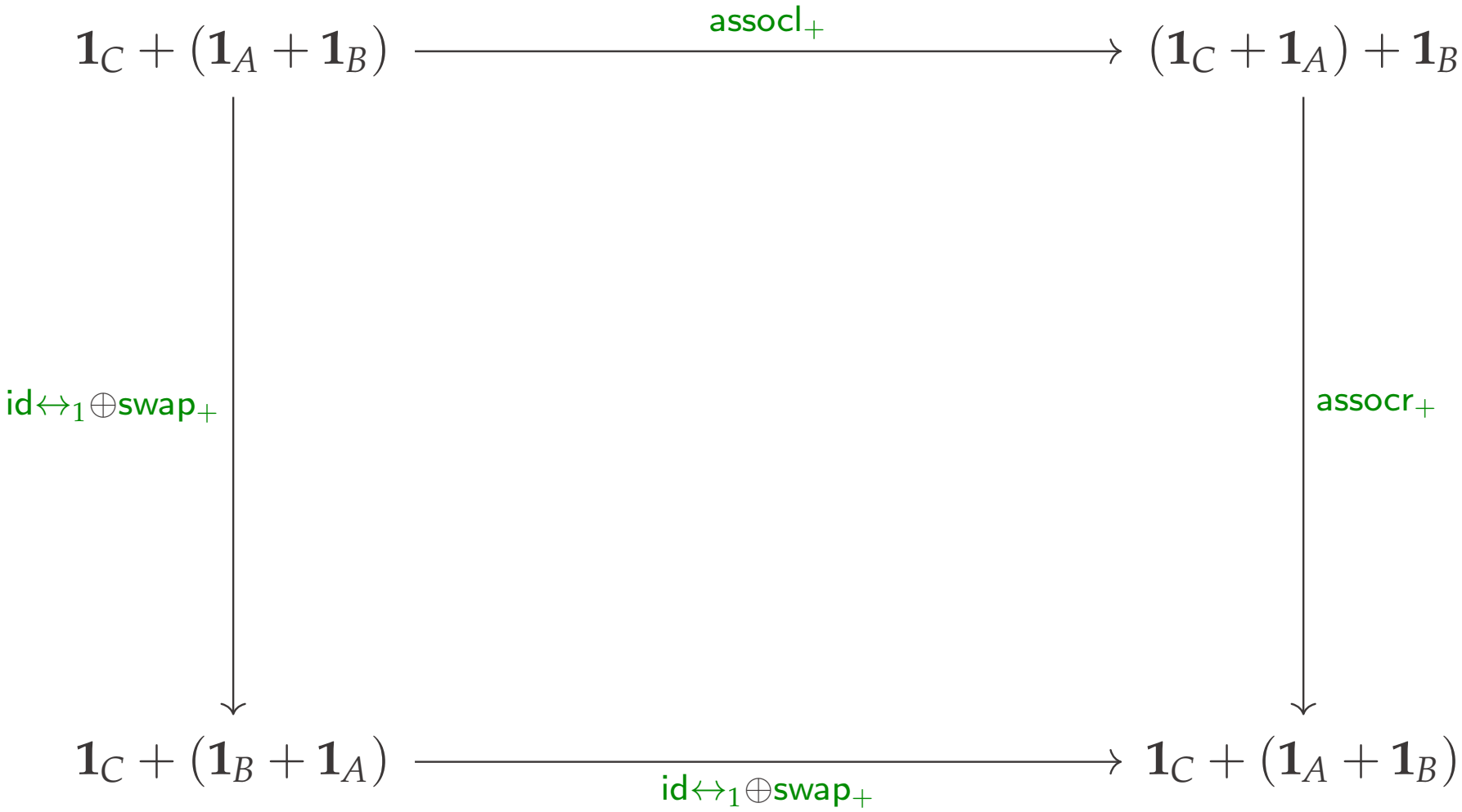
Hypothesis on the example



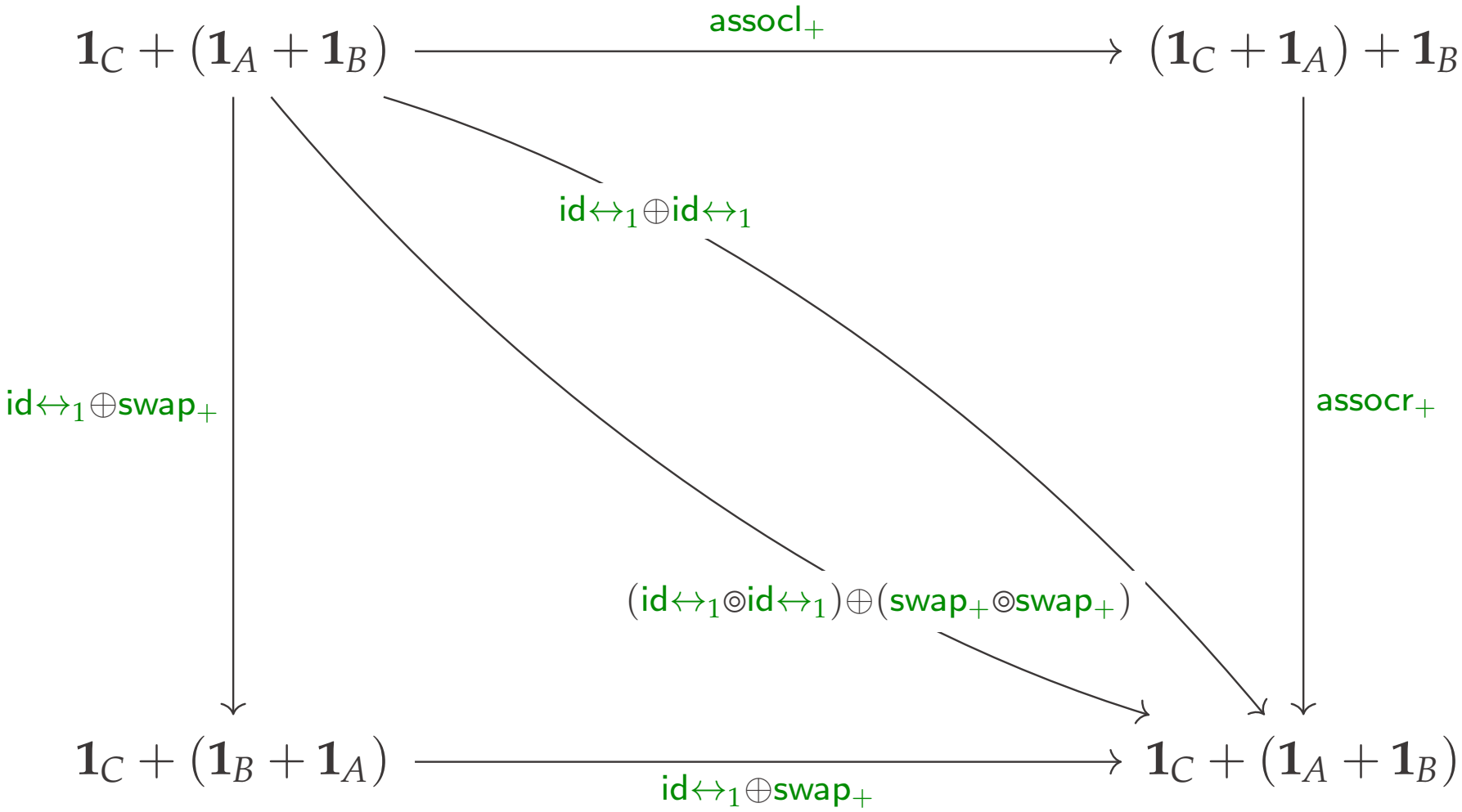
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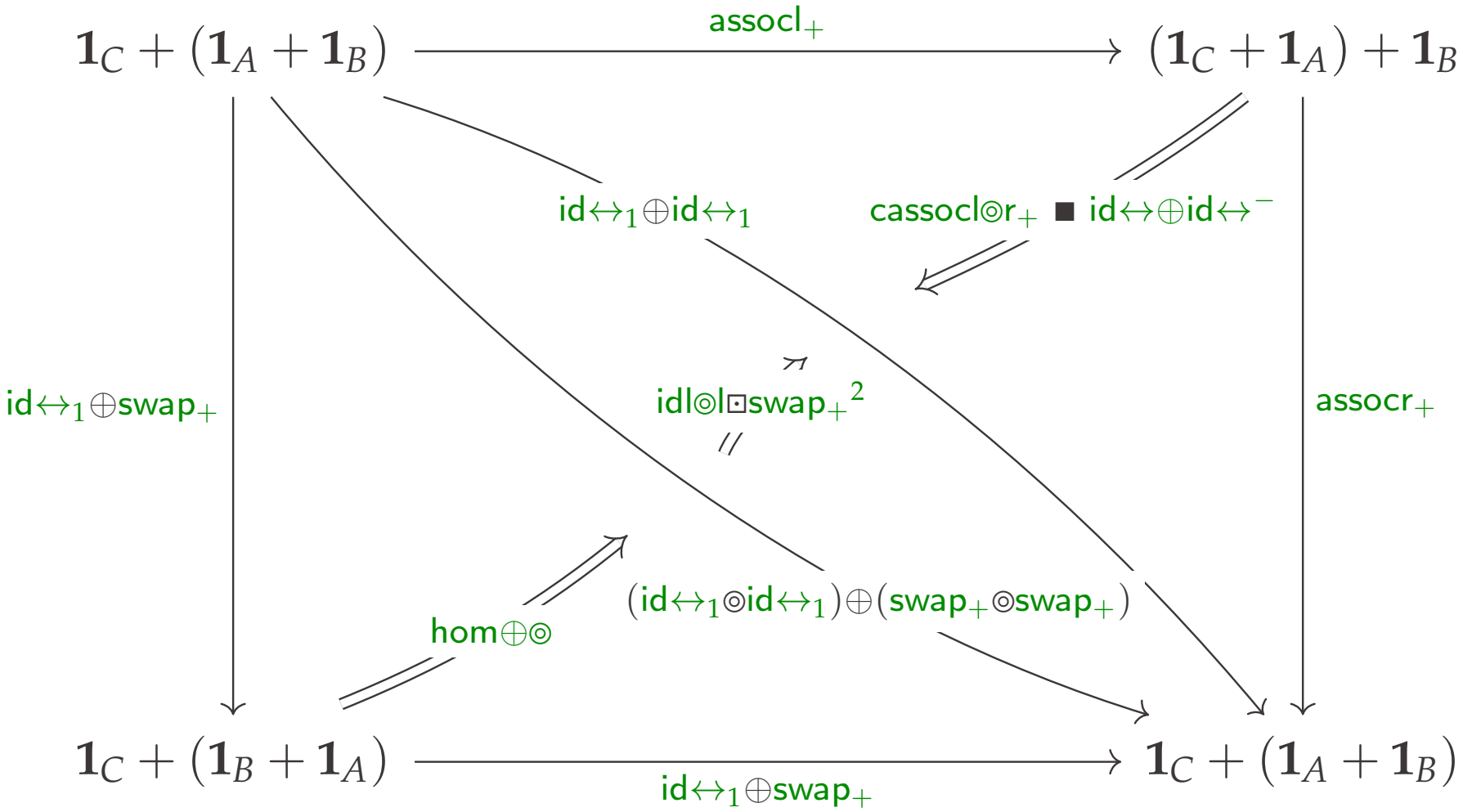
Hypothesis on the example



Hypothesis on the example



Hypothesis on the example



Full 2-combinator

$$
\begin{aligned}
&\mathsf{p12} : \mathsf{p1}\ \{A\}\ \{B\}\ \{C\} \leftrightarrow_2 \mathsf{p2}\\
&\mathsf{p12} = \mathsf{assocl}_+ \circledcirc \mathsf{swap}_+ \circledcirc \mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+\\
&\qquad\qquad \leftrightarrow_2 \langle\ \mathsf{assoc}\circledcirc\mathsf{l}\ \rangle\\
&\qquad (\mathsf{assocl}_+ \circledcirc \mathsf{swap}_+) \circledcirc \mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+\\
&\qquad\qquad \leftrightarrow_2 \langle\ ((\mathsf{idr}\circledcirc\mathsf{r} \blacksquare (\mathsf{id}\!\leftrightarrow_2 \boxdot \mathsf{linv}\circledcirc\mathsf{r}) \blacksquare \mathsf{assoc}\circledcirc\mathsf{l} \blacksquare (\mathsf{hexagonl}_+\mathsf{l} \boxdot \mathsf{id}\!\leftrightarrow_2)))\\
&\qquad\qquad\qquad \boxdot (\mathsf{idl}\circledcirc\mathsf{r} \blacksquare (\mathsf{linv}\circledcirc\mathsf{r} \boxdot \mathsf{id}\!\leftrightarrow_2))\ \rangle\\
&\qquad ((((\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+) \circledcirc \mathsf{assocl}_+) \circledcirc \mathsf{swap}_+ \oplus \mathsf{id}\!\leftrightarrow_1) \circledcirc \mathsf{assocr}_+)\\
&\qquad\qquad\qquad\qquad \circledcirc (\mathsf{id}\!\leftrightarrow_1 \circledcirc \mathsf{id}\!\leftrightarrow_1) \circledcirc \mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+\\
&\qquad\qquad \leftrightarrow_2 \langle\ (\mathsf{id}\!\leftrightarrow_2 \boxdot (\mathsf{linv}\circledcirc\mathsf{l} \boxdot \mathsf{id}\!\leftrightarrow_2)) \blacksquare (\mathsf{id}\!\leftrightarrow_2 \boxdot \mathsf{idl}\circledcirc\mathsf{l})\ \rangle\\
&\qquad ((((\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+) \circledcirc \mathsf{assocl}_+) \circledcirc \mathsf{swap}_+ \oplus \mathsf{id}\!\leftrightarrow_1) \circledcirc \mathsf{assocr}_+)\\
&\qquad\qquad\qquad\qquad \circledcirc \mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+\\
&\qquad\qquad \leftrightarrow_2 \langle\ \mathsf{assoc}\circledcirc\mathsf{r} \blacksquare \mathsf{assoc}\circledcirc\mathsf{r} \blacksquare \mathsf{assoc}\circledcirc\mathsf{r}\ \rangle\\
&\qquad (\mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+) \circledcirc \mathsf{assocl}_+ \circledcirc (\mathsf{swap}_+ \oplus \mathsf{id}\!\leftrightarrow_1) \circledcirc \mathsf{assocr}_+\\
&\qquad\qquad\qquad\qquad \circledcirc \mathsf{id}\!\leftrightarrow_1 \oplus \mathsf{swap}_+\\
&\qquad\qquad \leftrightarrow_2 \blacksquare
\end{aligned}
$$

Hypothesis

Hypothesis

Sound and complete list of 2-combinators for Π *is given by the coherence conditions for weak symmetric rig groupoids.*

Hypothesis

Hypothesis

Sound and complete list of 2-combinators for Π *is given by the coherence conditions for weak symmetric rig groupoids.*

Theorem

The free weak symmetric rig groupoid on zero generators is equivalent to the groupoid of finite sets and bijections.

Proof

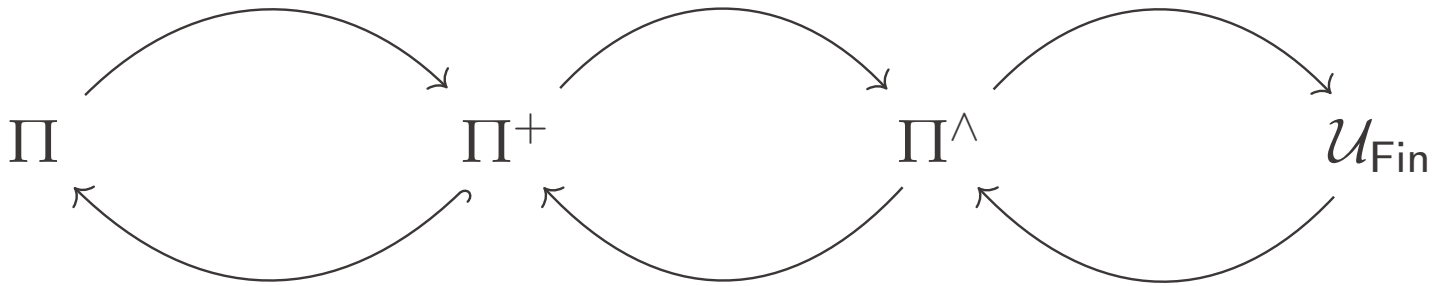
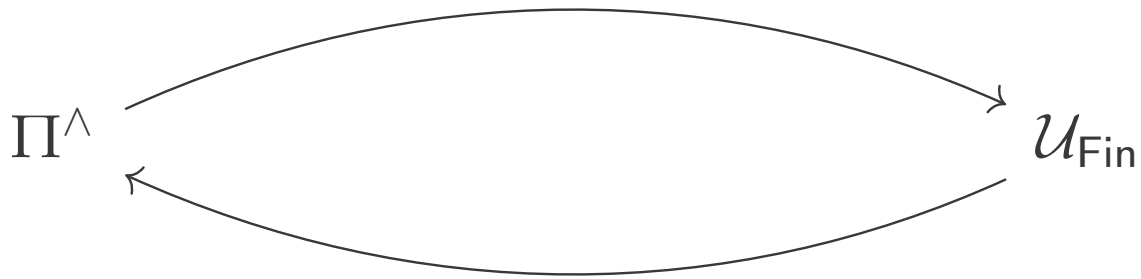


Figure: Proof in a few steps

$$(1+1)\times(1+1) \Leftrightarrow (1+1)+(1+1) \Leftrightarrow 1+(1+(1+(1+0))) \Leftrightarrow \mathsf{Fin}_4$$

Figure: Proof on types

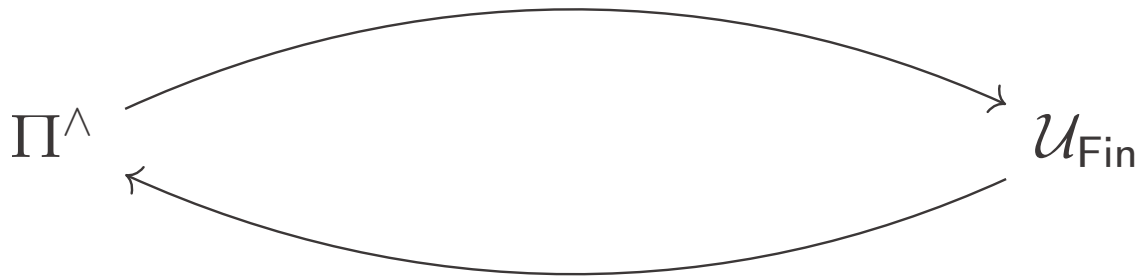
Proof - collapse



$(\mathsf{id}\leftrightarrow_1 \circledcirc \mathsf{id}\leftrightarrow_1 \circledcirc \mathsf{id}\leftrightarrow_1) : \mathbf{1} \leftrightarrow_1 \mathbf{1}$

$\{\star \mapsto \star\} : \mathsf{Fin}_1 \simeq \mathsf{Fin}_1$

Proof - collapse



$(\mathsf{id}\leftrightarrow_1 \circledcirc \mathsf{id}\leftrightarrow_1 \circledcirc \mathsf{id}\leftrightarrow_1) : \mathbf{1} \leftrightarrow_1 \mathbf{1}$

$\mathsf{id}\leftrightarrow_1 : \mathbf{1} \leftrightarrow_1 \mathbf{1}$

$\{\star \mapsto \star\} : \mathsf{Fin}_1 \simeq \mathsf{Fin}_1$

Proof

Technical devices:

- ⋆ Term rewriting system
- ⋆ Coxetera presentations of $\mathcal{S}_n$ groups
- ⋆ Lehmer codes
- ⋆ ...

Formalization

- ⋆ Everything in a proof assistant
- ⋆ Formal system: *Homotopy Type Theory*

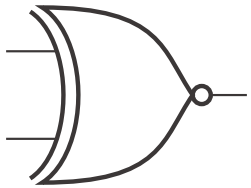
The end

Part of the thesis published as:

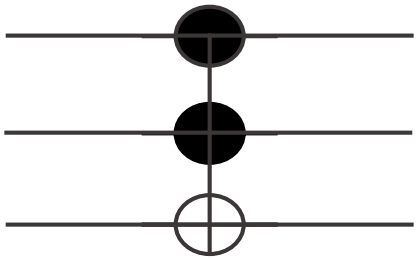
Vikraman Choudhury, Jacek Karwowski, and Amr Sabry. 2022. Symmetries in reversible programming: from symmetric rig groupoids to reversible programming languages. Proc. ACM Program. Lang. 6, POPL, Article 6 (January 2022), 32 pages. https://doi.org/10.1145/3498667

Appendix A

Reversible logic circuits



NAND is a universal logic gate



Toffoli is a universal *reversible* logic gate

Reversible circuit equivalence problem

Theorem

Reversible circuit equivalence problem is co-NP complete.

Hypothesis

Reversible circuit equivalence problem is in NP.

Hypothesis

co-NP $\subset$ *NP*

Reversible circuit equivalence problem

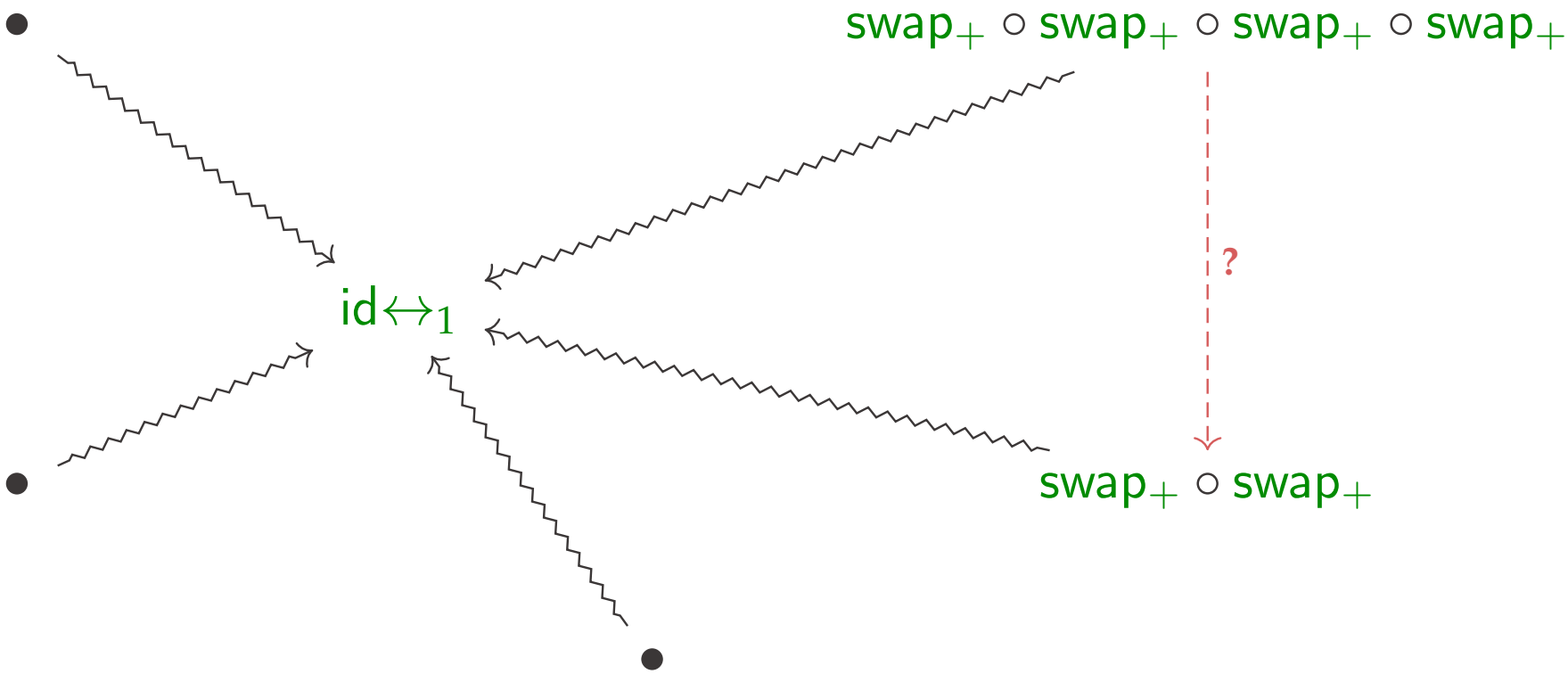
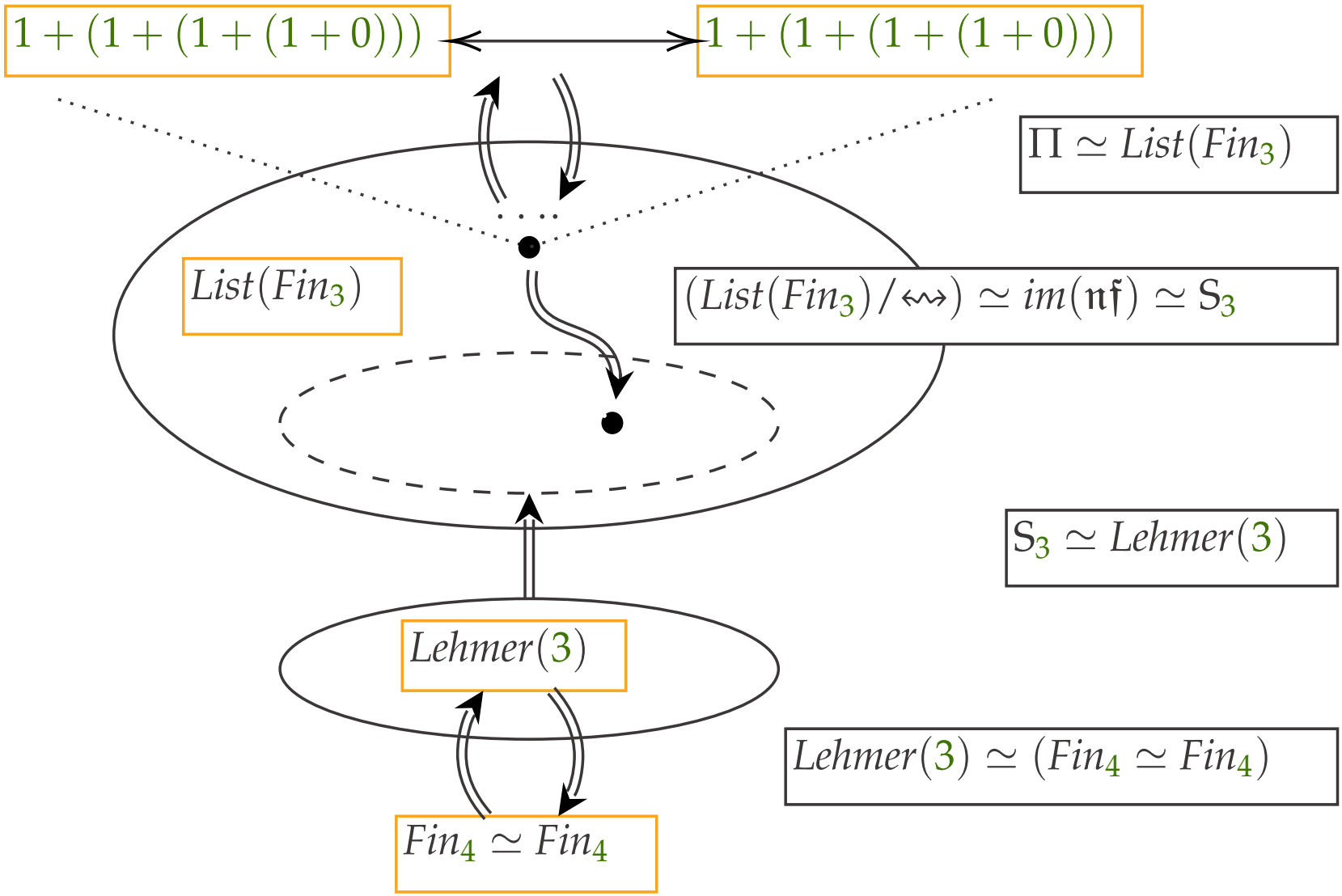


Figure: Structure of the reversible circuit equivalence graph

Appendix B

Proof - a bird's eye view

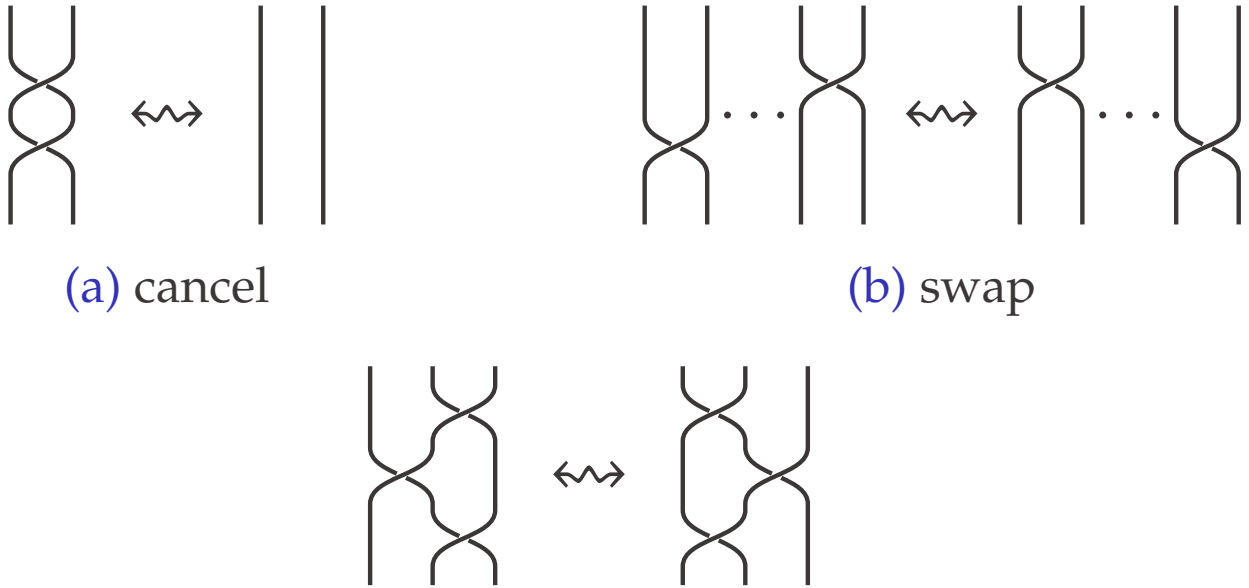


Proof - term rewriting system

Theorem

$$(List(Fin_n)/\leftrightsquigarrow) \simeq \mathcal{S}_n \simeq Lehmer(n) \simeq (Fin_{\mathsf{S}\,n} \simeq Fin_{\mathsf{S}\,n})$$

Proof - normalization



(a) cancel

(b) swap

(c) braid

Figure: First idea for $\rightsquigarrow$: directing the Coxeter presentation

$$[\ldots, \tau_k, \tau_k \ldots] \rightsquigarrow [\ldots, \ldots] \qquad (cancel)$$

$$[\ldots, \tau_{m+k+2}, \tau_m \ldots] \rightsquigarrow [\ldots, \tau_m, \tau_{m+k+2}, \ldots] \qquad (swap)$$

$$[\ldots, \tau_{m+1}, \tau_m, \tau_{m+1} \ldots] \rightsquigarrow [\ldots, \tau_m, \tau_{m+1}, \tau_m, \ldots] \qquad (braid)$$

Proof - normalization

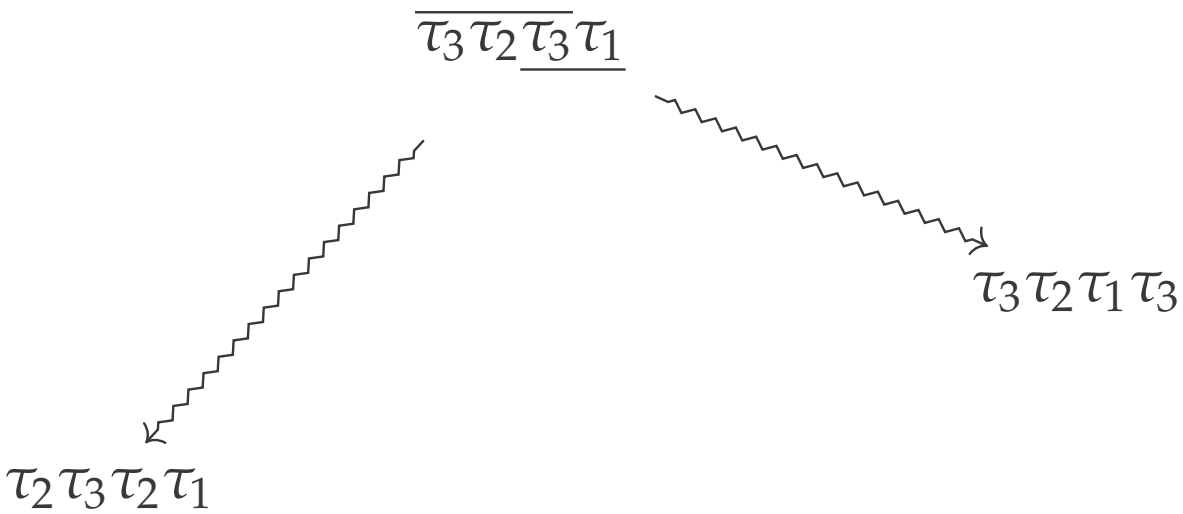


Figure: Critical pairs does not converge!

Proof - new equations

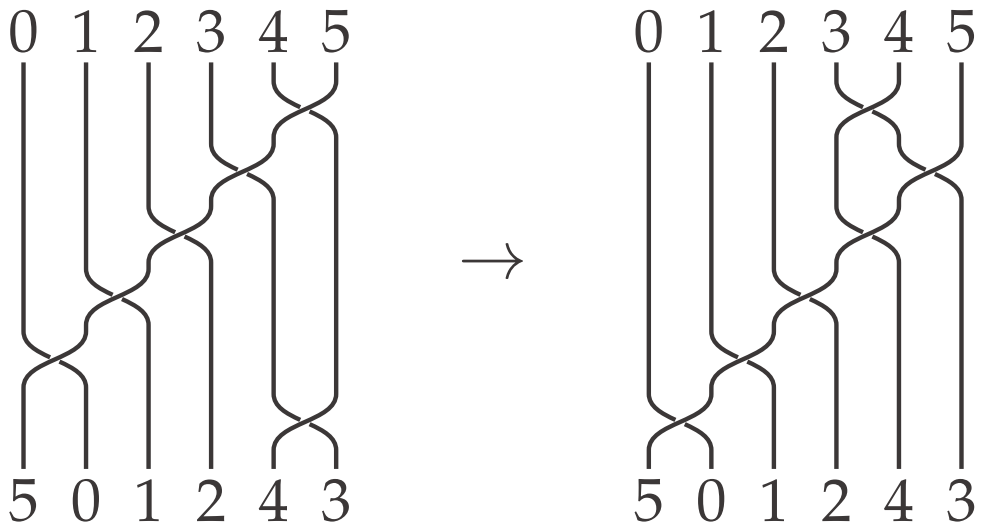


Figure: Long braid

$$[\tau_4, \tau_3, \tau_2, \tau_1, \tau_0, \tau_4] \rightsquigarrow [\tau_3, \tau_4, \tau_3, \tau_2, \tau_1, \tau_0]$$

Term rewriting system

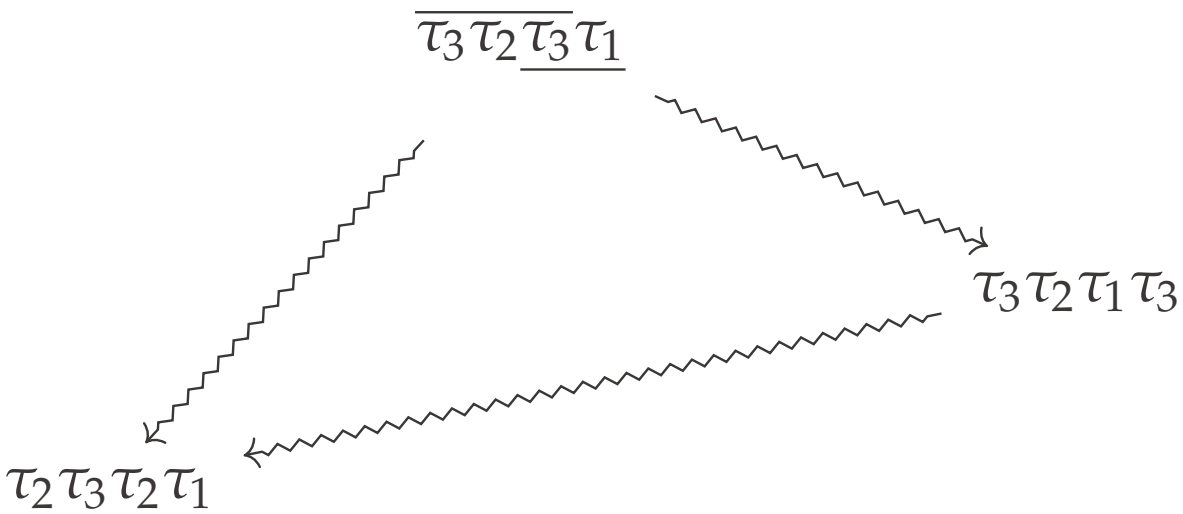


Figure: In the new system, all critical pairs converge.

How do the $\rightsquigarrow$ normal forms look like?

Lehmer codes

Definition

$$\begin{aligned} \mathit{Lehmer}(0) &= \mathrm{Fin}_{\mathsf{S}\,0} \\ \mathit{Lehmer}(\mathsf{S}\,n) &= \mathit{Lehmer}(n) \times \mathrm{Fin}_{\mathsf{S}\,\mathsf{S}\,n} \end{aligned}$$

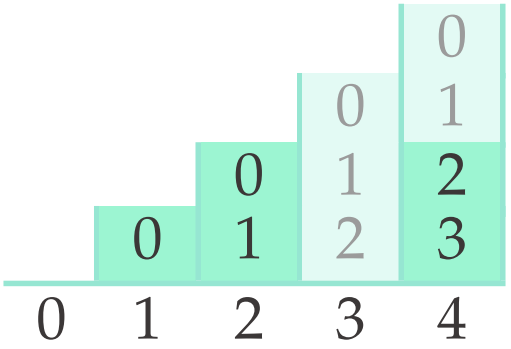
In other words:

$$\begin{aligned} \mathit{Lehmer}(0) &= \mathrm{Fin}_1 \\ \mathit{Lehmer}(1) &= \mathrm{Fin}_1 \times \mathrm{Fin}_2 \\ \mathit{Lehmer}(2) &= \mathrm{Fin}_1 \times \mathrm{Fin}_2 \times \mathrm{Fin}_3 \\ \mathit{Lehmer}(3) &= \mathrm{Fin}_1 \times \mathrm{Fin}_2 \times \mathrm{Fin}_3 \times \mathrm{Fin}_4 \\ &\dots \end{aligned}$$

Lehmer codes are the normal forms

For example:

$$(0,1,2,0,2) : \mathrm{Fin}_1 \times \mathrm{Fin}_2 \times \mathrm{Fin}_3 \times \mathrm{Fin}_4 \times \mathrm{Fin}_5 = \mathit{Lehmer}(4)$$



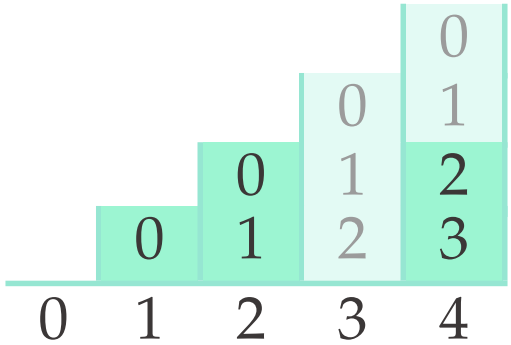
Representing Lehmer code as a sequence of adjacent transpositions:

$$[0,1,0,3,2] : \mathit{List}(\mathrm{Fin}_4)$$

Lehmer codes are the normal forms

For example:

$$(0,1,2,0,2) : \mathrm{Fin}_1 \times \mathrm{Fin}_2 \times \mathrm{Fin}_3 \times \mathrm{Fin}_4 \times \mathrm{Fin}_5 = \mathit{Lehmer}(4)$$



Representing Lehmer code as a sequence of adjacent transpositions:

$$[0,1,0,3,2] : \mathit{List}(\mathrm{Fin}_4)$$
$$[0,1,0,2,1,0,3,2,1,0] : \mathit{List}(\mathrm{Fin}_4)$$